

Comparative Evaluation of Transformer Models for AI-Generated Text Detection in English and French

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Abstract :

This manuscript provides a multilingual approach to classify AI-generated text and human-written text in English and French, through the utilization of transformer-based models, which allow us to capture deep semantic and stylistic distinctions between human authors versus machine authors, we were able to tune the decision threshold on a training set using the validation set in order to find balanced precision versus recall, and our best performing model, a distilled version of Multilingual BERT trained using Efficient Fine Tuning, demonstrated excellent language independence and classification consistency across the two languages, thereby showing the strength of multilingual transformers and their importance for developing dependably balanced systems for the detection of Artificial intelligence.

keywords : AI-generated text detection, multilingual natural language processing, English–French text classification, human vs machine classification, transformer-based models, threshold optimization, parameter-efficient fine-tuning

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The code associated with this study is available on GitHub: <https://github.com/ae-saouiqui/ai-text-detection>

I. Introduction

The rapid advancement of large language models has enabled the generation of highly coherent and human-like text. While these developments have created new opportunities in domains such as education, content creation, and automated assistance, they have also introduced significant challenges. One of the most critical issues is the growing difficulty of distinguishing between human-written and AI-generated text, as modern generative systems increasingly produce outputs that closely resemble natural human language. This challenge becomes even more complex in multilingual environments, where differences in syntax, vocabulary, and linguistic structure may influence both model behavior and detection performance. In particular, detecting AI-generated content across languages such as English and French requires approaches that are robust to linguistic diversity. Consequently, multilingual AI-generated text detection has emerged as an active and important research area.

This challenge has motivated a growing body of research focused on the detection of machine-generated content, often evaluated through shared tasks and benchmark datasets

covering both monolingual and multilingual settings (Wang et al., 2024). These studies highlight the complexity of the problem, as detection models must generalize across diverse domains, languages, and generative sources. However, existing approaches still exhibit important limitations, particularly under domain shift and adversarial conditions such as paraphrasing, where detection performance often degrades significantly (Antoun et al., 2023). Similar observations have been reported for language-specific settings such as French, where models may achieve strong results on controlled datasets but struggle to generalize beyond the training distribution. These challenges highlight the need for more robust and generalizable approaches to multilingual AI-generated text detection in real-world scenarios. In this work, we focus on a multilingual approach for AI-generated text detection, targeting both English and French texts.

In this work, we investigate a multilingual framework for AI-generated text detection targeting both English and French texts. The objective is to develop a robust classification system capable of distinguishing human-written content from machine-generated text using transformer-based representations, which are well suited for capturing semantic and

contextual information across multiple languages. To improve robustness and generalization, particularly for French where existing approaches often rely on limited-domain datasets, we incorporate additional heterogeneous sources including essays, conversational data, and code-generation samples. Furthermore, special attention is given to balanced detection performance across both classes through threshold optimization on the validation set, allowing a better trade-off between precision and recall. The final evaluation assesses the generalization capability of the selected models on a fully held-out test set.

The remainder of this paper is organized as follows. Section 2 reviews related work and prior studies in the field. Section 3 describes the datasets and data preparation process. Section 4 presents the proposed methodology, including the training procedure, model architectures, threshold optimization strategy, and experimental results. Finally, Section 5 concludes the paper.

II. Related Work

This section presents previous studies related to AI-generated text detection, focusing on approaches that are relevant to our work.

A survey by Xiang et al. reviews recent progress in AI-generated text detection and proposes a taxonomy relevant to our work. They categorize methods into passive and active approaches, depending on whether they rely on intrinsic textual features or embedded signals. Passive methods include surface linguistic and language model-based techniques, while active methods involve watermarking and semantic retrieval strategies. Their study also highlights key limitations of current detectors, particularly their vulnerability to adversarial attacks such as paraphrasing, which significantly reduces performance and generalization. Overall, their findings emphasize the need for more robust and domain-general detection methods, which aligns with the goals of our multilingual approach [1].

Adilazuarda et al. conduct a comprehensive comparison of traditional shallow classifiers, language model fine-tuning, and multilingual model fine-tuning for machine-generated text detection and model attribution. Using FastText-based features and multiple pre-trained transformers, they show that fine-tuned pre-trained language models substantially outperform shallow methods and that multilingual fine-tuning (mBERT, DistilBERT) helps mitigate language-specific biases and maintain strong performance across

languages, highlighting the importance of multilingual training strategies for robust detection [2].

Antoun et al. investigate ChatGPT detection in multilingual and cross-domain settings. They train classifiers using translated datasets and evaluate them in both monolingual and multilingual scenarios. Their study further examines generalization on out-of-domain data and evaluates robustness against adversarial attacks such as character substitutions and misspellings. Their results reveal important limitations in existing detectors, particularly in real-world and adversarial conditions, emphasizing the need for more robust and generalizable approaches [3].

Wang et al. introduce the GenAI Content Detection Task, a shared benchmark designed to evaluate machine-generated text detection in both monolingual and multilingual settings. The task includes diverse domains, multiple state-of-the-art generative models [Figure B.B.1], and a wide range of languages, making it a challenging benchmark for evaluating generalization. Their work provides a comprehensive evaluation framework that reflects real-world conditions and highlights the difficulty of robust detection across languages, domains, and modern generative systems [4].

III. Datasets

1. Dataset collection :

Several benchmark datasets have been proposed for machine-generated text detection. Early efforts such as HC3 (Guo et al., 2023) include both English and Chinese texts generated by large language models. Other datasets, including MGTBench, ArguGPT, and DeepfakeTextDetect, focus on texts generated by different large language models across various domains. More comprehensive benchmarks such as M4 and M4GT-Bench (Wang et al., 2024) extend this work by covering multiple domains, languages, and generation sources. [Tableau A.1]

In our study we focus only on the English portion of these benchmarks [5]. For the French dataset, it is constructed from two main sources. The first is the HC3-OOD dataset provided by Antoun et al. [6], while the second is a French instruction dataset composed of multiple translated and native French corpora [7]. The HC3-OOD dataset contains human and ChatGPT-generated conversations adapted to out-of-domain settings, providing challenging examples for detection. The French instruction dataset includes a mixture of translated instruction-following datasets and native French sources such as WikiHow FR and other conversational

corpora. This dataset was originally designed for training and evaluating instruction-based language models, but it can also be used for distinguishing between human-written and AI-generated text [Tableau A.2]

Fortunately, both provided a test set drawn from different sources than the training set, serving as a valid benchmark for model generalization .

Table 1 and *Table 2* summarizes respectively the number of samples in the English and French datasets aimed to used for training in this work.

Table 1 : Summary of Training Dataset Samples

Dataset	English	French
Count	610767	464885
Total	1075652	

Table 2 : Summary of Test Dataset samples

Dataset	English	French
Count	70000	15000
Total	85000	

2. Dataset preparation :

The collected data underwent several preprocessing steps to improve consistency and quality, including the removal of hyperlinks, markdown syntax, and other formatting artifacts, followed by standard text

preprocessing. For model training, a subset of 500,000 samples was selected, consisting of 200,000 English and 200,000 French samples, each maintaining a balanced distribution between human-written and AI-generated text.

To improve robustness against noisy, real-world inputs, an additional 100,000 augmented samples were introduced containing OCR and keyboard-induced perturbations [8]. This augmented subset was designed with asymmetric class distributions depending on the noise type: OCR-based noise contains 80% AI-generated and 20% human-written samples, while keyboard-error augmentation follows the inverse distribution (80% human-written, 20% AI-generated). In addition to the training data, a separate validation set of 100,000 samples was constructed with an imbalanced distribution (80% human-written, 20% AI-generated) to better reflect realistic scenarios where human-authored content is more prevalent.

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IV. Methodology

1. Model Architecture

The proposed approach is based on transformer encoder models adapted for binary sequence classification, where each model consists of a pre-trained transformer encoder followed by a task-specific classification head that outputs the probability of a text being human-written or AI-generated [\[Figure 1\]](#). To evaluate the effectiveness of different pre-trained representations, multiple

transformer-based backbones are fine-tuned within the same classification framework, and despite differences in their pre-training strategies and architectures, all models share the same overall design and are adapted to the same downstream task. The framework is designed to support multilingual inputs, enabling consistent processing of both English and French texts within a unified classification setup.

Through the application of the Hugging Face Trainer API, we will conduct experiments on fine-tuning transformers via the application of LoRA (low-rank adaptation) for parameter-efficient adaptation. After fine-tuning, we will optimize our thresholds through a validation-based approach so that we can improve the classification balance between text written by humans and text generated by AI. The evaluation process will involve computing several classification performance metrics for the previously mentioned models, including precision, recall, F1 score and accuracy, which are found in [\[Appendix F — Evaluation Metrics\]](#). These metrics will provide insight into how well each of the fine-tuning methods and threshold optimization methods

performed in classifying AI-generated text.

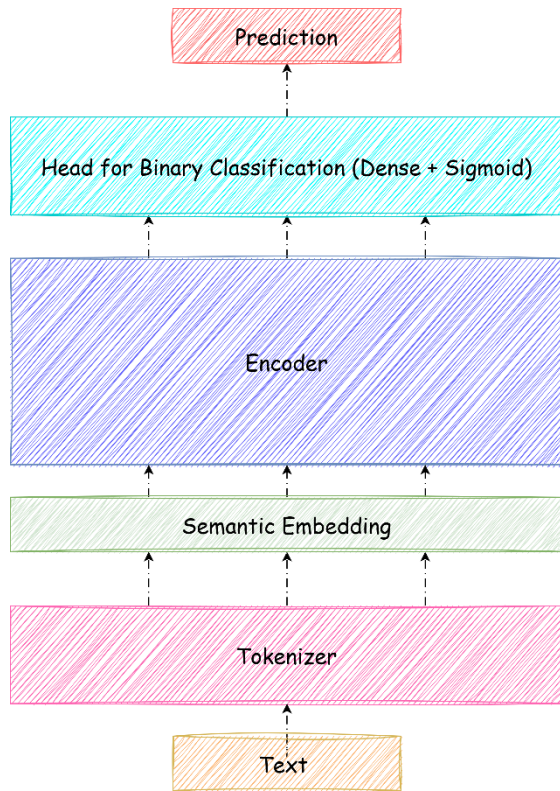


Figure 1 : Model Architecture

2. Fine-tuning Strategy :

To evaluate different transformer architectures and fine-tuning strategies, we conduct experiments using three configurations: mELECTRA (Clark et al.2020) [9], mDistilBERT (Sanh et al., 2020) [10] , and mDistilBERT with Low-Rank Adaptation (LoRA) (Hu et al., 2021) [11] .

All models are adapted for binary sequence classification using a shared classification framework consisting of a transformer encoder followed by a classification

head [Figure 1].The first two models (mELECTRA and mDistilBERT) are fine-tuned using standard full-parameter training.

In contrast, the third configuration applies LoRA to mDistilBERT, where only a small number of additional low-rank parameters are trained while the original backbone weights remain frozen. This allows for a comparison between full fine-tuning and parameter-efficient fine-tuning under identical training conditions. This setup enables a fair evaluation of both architectural differences and the impact of LoRA on model performance for AI-generated text detection.

3. Hyperparameter configuration :

All models are fine-tuned using the Hugging Face Trainer API [12] , which provides a unified training framework for transformer-based sequence classification. This ensures consistent implementation across all experiments, including data loading, optimization, evaluation, and checkpoint management. The main hyperparameters used for each model configuration are summarized in *Table 3* .

Table 3 : Hyperparameters used for finetuning

Parameter	mELECTRA	mBERT-Dis-til	mBERT-Distil + LoRA
Epochs	3	3	2
Batch Size	4	4	16
Learning Rate	2×10^{-5}	2×10^{-5}	2×10^{-4}
Weight Decay	0.01	0.01	0.01
Max Sequence Length	dynamic < 512	dynamic < 512	512
Optimizer	AdamW	AdamW	AdamW
Warmup Steps	—	—	500
Scheduler	—	—	Cosine
Mixed Precision	FP16	FP16	FP16

In our setup, for the mDistilBERT model, we apply Low-Rank Adaptation (LoRA) to enable parameter-efficient fine-tuning which is applied specifically to the query and value projection matrices of the self-attention mechanism. The configuration used is as follows in *Table 5* :

Table 4 : LoRA configuration

Parameter	Value
Task Type	Sequence Classification
Rank (r)	8
LoRA Alpha	16
Dropout	0.1
Bias	None
Target Modules	query, value

4. Threshold optimization :

After Training, we perform threshold optimization on the validation set to improve classification reliability. The optimal threshold is selected by evaluating different cutoff values and choosing the one that maximizes and compromises both precision and recall across the two classes. This approach ensures a balanced trade-off between false positives and false negatives, which is particularly important in AI-generated text detection where both error types are equally critical.

The validation results of the ELECTRA model before threshold optimization [Table 5] reveal a clear imbalance in prediction behavior. While the model achieves high recall for the

AI-generated class (0.95), its precision remains relatively low (0.57), indicating a tendency to over-predict AI-generated text. In contrast, the human class exhibits high precision (0.98) but lower recall (0.82), suggesting that a portion of human-written samples are incorrectly classified as AI-generated. These results highlight the limitation of using a default decision threshold of 0.5, motivating the need for threshold optimization to achieve a more balanced trade-off between precision and recall across both classes.

Table 5 : classification report of mElectra on validation set with a threshold of 0.5

	precision	recall	f1-score
Human	0.98	0.82	0.9
AI	0.57	0.95	0.71
Accuracy	0.85		

The decision threshold is selected based on validation performance with a focus on improving detection quality for the AI-generated class.

Specifically, we choose the threshold that ensures a minimum performance level (approximately 80%) for both precision and recall of the AI class, as this class represents the main source of classification errors in the default setting. This constraint helps reduce misclassification of AI-generated

text while maintaining a balanced overall performance which has been shown in the [Figure 4].

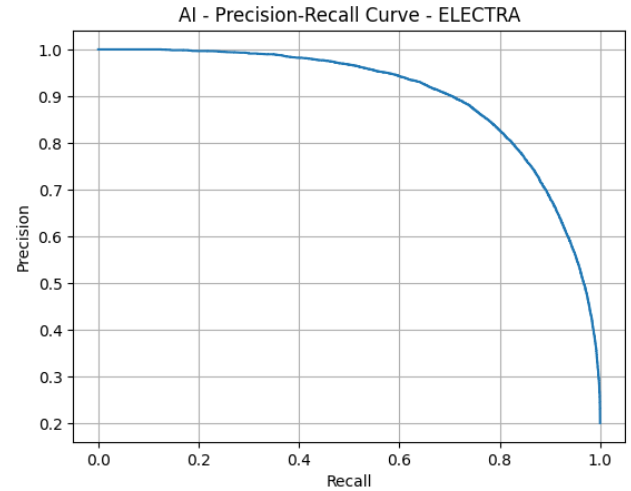


Figure 4: AI Precision - Recall curve for Electra

After applying the optimal decision threshold selected on the validation set, mELECTRA achieves strong and balanced performance as shown in table 7. The model reaches an overall accuracy of 0.93, with a macro-level improvement in class separation compared to the initial default threshold setting. Specifically, the Human class is classified with high reliability, obtaining a precision of 0.95, recall of 0.96, and F1-score of 0.95. For the AI class, the model achieves a precision of 0.83, recall of 0.80, and an F1-score of 0.81, indicating a good trade-off between correctly identifying AI-generated text and minimizing misclassifications. Overall, these results confirm that threshold optimization improves the balance between both classes, particularly enhancing the detection of the

more challenging AI-generated category.

Table 6 : classification report of mElectra on validation set with optimal threshold

	precision	recall	f1-score
Human	0.95	0.96	0.95
AI	0.83	0.80	0.81
Accuracy	0.93		

The same threshold optimization strategy is applied to all remaining models, where the decision threshold is selected based on validation performance and guided by the class exhibiting the highest misclassification rate under the default setting, ensuring a balanced trade-off between precision and recall based on the plots in Figure and Figure .

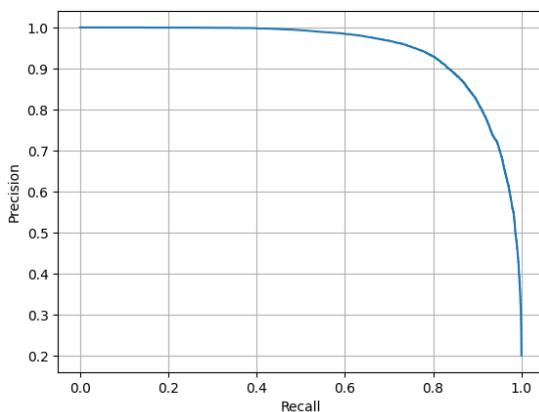


Figure 5: AI Precision - Recall curve for mDistill-BERT

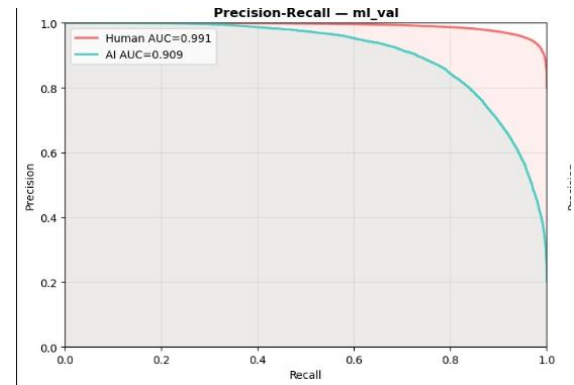


Figure 6 : AI-Human Precision - Recall curve for mDistill-BERT + LoRA

The table 8 reports thresholds chosen for each model after analysing corresponding curve for each model .

Table 7 : Thresholds chosen after analysis

Model	Threshold / Class
mELECTRA	0.96 / AI
mBERT-Distil	0.99 / AI
mBERT-Distil + LoRA	0.0215 / Human

Following the selection of the optimal decision threshold on the validation set, all three models demonstrate strong and consistent performance, with overall accuracies ranging from 0.93 to 0.95, as reported in Table 9. In particular, mBERT-Distil achieves the highest accuracy (0.95) and the strongest overall balance, with an F1-score of 0.86 for the AI class and 0.97 for the Human class. mELECTRA and mBERT-Distil + LoRA also show competitive results, achieving stable Human-class performance (F1-scores of 0.95 and 0.96 respectively) while maintaining solid AI detection capability. Across all models, the Human class is consistently well recognized, whereas the AI class remains more challenging, with slightly lower recall values. Overall, these results indicate that threshold optimization leads to well-balanced classification behavior and reinforces the robustness of all evaluated architectures.

Table 8 : evaluation scores on validation set after score optimization

Model	Accuracy	Precision (AI)	Recall (AI)	F1 (AI)	Precision (Human)	Recall (Human)	F1 (Human)
mELECTRA	0.93	0.83	0.80	0.81	0.95	0.96	0.95
mBERT-Distil	0.95	0.93	0.80	0.86	0.95	0.98	0.97
mBERT-Distil + LoRA	0.93	0.87	0.77	0.82	0.94	0.97	0.96

5. Test evaluation :

To evaluate the generalization capability of the proposed models, we report the final performance on the held-out test set as demonstrated in table 10. The decision thresholds selected on the validation set are kept fixed during testing to ensure a fair evaluation.

Table 9 : Final test performance :

Model	Accu- racy	Preci- sion (AI)	Recall (AI)	F1 (AI)	Precision (Human)	Recall (Hu- man)	F1 (Hu- man)
mELECTRA	0.78	0.77	0.85	0.81	0.80	0.69	0.74
mBERT-Distil	0.79	0.79	0.85	0.82	0.80	0.72	0.76
mBERT-Distil + LoRA	0.80	0.79	0.87	0.82	0.82	0.72	0.76

Futhermore. We evaluate the AI-class performance separately on English [table 11] and French [table 12] test sets to analyze the cross-lingual robustness of the proposed models.

Table 10 : test performance on English distribution

Model	Precision (AI)	Recall (AI)	F1 (AI)	Accuracy
mELECTRA	0.72	0.84	0.78	0.75
mBERT-Distil	0.74	0.84	0.79	0.76
mBERT-Distil (LoRA)	0.74	0.86	0.8	0.78

Table 11 : test performance on English distribution

Model	Precision (AI)	Recall (AI)	F1 (AI)	Accuracy
mELECTRA	0.97	0.88	0.92	0.90
mBERT-Distil	0.98	0.88	0.93	0.90
mBERT-Distil (LoRA)	0.97	0.87	0.92	0.89

The results show that all models achieve stable and competitive performance on the test set, confirming their generalization ability. mDistilBERT with LoRA achieves the best overall accuracy (0.80), demonstrating the effectiveness of parameter-efficient fine-tuning. Across all models, recall for AI-generated text remains consistently high, indicating strong detection capability, while performance on the human class is slightly lower in recall, reflecting a mild bias toward the AI class that persists even after threshold optimization.

V. Limitations and future perspectives

Despite achieving competitive results, several limitations remain. A noticeable performance gap

between the validation set (≥ 0.93) and the fully held-out test set (≤ 0.80) indicates sensitivity to domain shift and suggests that the models do not fully generalize to data outside the training distribution. This limitation is particularly visible for the English setting, where performance is lower than for French. One possible explanation is the greater diversity and heterogeneity of the English test data, which likely introduces more challenging out-of-distribution samples.

Another limitation of this study is the restricted multilingual scope, as experiments are conducted only on English and French datasets. Although the models demonstrate promising multilingual performance, the results may not generalize to other languages with different linguistic structures, writing systems, or resource availability. In addition, variations in dataset size and composition across languages may also influence performance behavior between the evaluated settings. Thirdly, the training phase did use basic noise augmentation from OCR and keyboards, and while no testing was done on the model against more complex adversarial transformations (paraphrased versions of sentences or

sentences translated back into the same language) we have evidence of significant disruption of overall performance when these tests were conducted on earlier models.

A further limitation of this work is the absence of probability calibration. Although the models produce confidence scores for classification, these outputs are not explicitly calibrated to reflect true likelihood estimates. As a result, the predicted probabilities may be overconfident or underconfident, which can reduce the reliability of the confidence scores, especially under distribution shifts across different datasets or domains. Incorporating calibration techniques such as temperature scaling could improve the interpretability and reliability of the model predictions.

Finally, the training data has relatively limited coverage of generative sources and may affect the model's ability to identify model outputs from new language models or language models that were not present in the original training data, to detect outputs from current generation language models.

This research has several limitations that could be overcome and expanded upon. The first is cross-domain generalization, which

could potentially improve with the implementation of domain-adversarial training, various data augmentation strategies, or ensemble methods that use multiple classifiers from different domains. Another area for further development is the application of this framework to other languages, in particular low-resource languages, via the use of massively multilingual models like XLM-RoBERTa. Additionally, establishing adversarial robustness as an active training objective, rather than an after-the-fact evaluation, is an important next step – especially for context-relevant detection where obfuscation is common. Implementing threshold calibration methods such as temperature scaling or Platt scaling (which are also referred to as dynamic thresholding methods) could help improve prediction accuracy by adapting to changing real-world distributions of observed data/outcomes. Finally, to ensure continued detection compatibility with emerging generative models, continual learning strategies can be implemented to refresh the model periodically with previously generated outputs. Lastly, using explanation tools, such as attention visualisation or probing classifiers (metrics

that indicate what words are most relevant to the classification) will shed light onto which feature types may be most relevant, allowing for improvement of the model and producing a basis for developing human-in-the-loop review during high-stakes deployment scenarios.

VI. Conclusion

In this work, We looked at how to detect AI-generated text in English and French, focusing on both single and multi-lingual examples. We came up with a standardised way to classify text with transformer-based models, and tried out different architectures (including mELECTRA, and mBERT-Distil) with an additional LoRA model for more efficient parameter fine-tuning. We also created a procedure for optimising threshold values based on the results of validation to provide us with an improved balance between precision and recall across the classes (where there was a systemic bias towards all AI-generated text when the default threshold was used). This has led to a more stable and consistent set of results from the above-mentioned models. Our experimental results demonstrate that all models produced comparable results on the held-out test set (with mBERT-Distil + LoRA providing the most consistent performance), while

reducing the total number of learnable parameters significantly. In addition, the evaluation of models across both English and French indicates that this method has reasonable performance in both languages in the presence of the challenges posed by both a cross-lingual environment and a domain variable.

In conclusion, the multi-lingual transformer models with parameter-efficient fine-tuning combined with threshold calibration prove to be an effective method for AI-generated text detection. Future work includes exploring how well the models perform in other domains (i.e., domains very different from those which they were trained) as well as testing the model's resistance to adversarial attacks (e.g., paraphrasing). The study also provides future research opportunities regarding additional languages and new AI-generative models.

VII. References

- [1] X. Lingyun, N. Lil, Y. Liu and J. Hu, "AI-Generated Text Detection: A Comprehensive Review of Active and Passive," *Computers, Materials & Continua*, p. 29, 2026. [[Link](#)]
- [2] M. Adilazuarda, "Beyond Turing: A Comparative Analysis of Approaches for Detecting," arXiv, Zagreb, 2024. [[Link](#)]
- [3] W. Antoun, "Towards a Robust Detection of Language Model-Generated Text:," Paris, 2024. [[Link](#)]
- [4] Y. Wang, "GenAI Content Detection Task 1: English and Multilingual," 2025. [[Link](#)]
- [5] Y. Wang, "English dataset : Jinyan1/COLING_2025_MGT_en" [[Link](#)].
- [6] W. Antoun, "French Dataset-1 : almanach/hc3_french_ood." [[Link](#)].
- [7] "French Dataset-2 : angeluriot/french_instruct." [[Link](#)].
- [8] "NLP augmentation tool" [[Link](#)].
- [9] K. Clark, "ELECTRA: PRE-TRAINING TEXT ENCODERS AS DISCRIMINATORS RATHER THAN GENERATORS" arXiv, 2020. [[Link](#)]
- [10] V. Sanh, "DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter," 2019. [[Link](#)]
- [11] E. Hu, "LORA: LOW-RANK ADAPTATION OF LARGE LAN," 2021. [[Link](#)]
- [12] "Fine-tuning a model with the Trainer API" [[Link](#)].
- [13] Evaluating the Performance of AI Text Detectors, Few-Shot and Chain-of-Thought Prompting Using DeepSeek Generated Text [[Link](#)].
- [14] CEALD: Benchmark of Multilingual Machine-Generated Text Detection Methods for Central European Languages [[Link](#)].
- [15] GPTZero: Robust Detection of LLM-Generated Texts [[Link](#)]
- [16] Unsupervised Cross-lingual Representation Learning at Scale [[Link](#)].

[17] [Parameter-Efficient Fine-Tuning Techniques for Transformer Models](#) [[Link](#)].

[18] [Attention Is All You Need](#) [[Link](#)].

VIII. Appendix

A. Appendix A – Dataset Statistics:

Tableau A.1 : English Dataset sources

Split	Sources	Total
Train	HC3, M4GT, MAGE	610,767
Test	CUDRT, IELTS, NLPeer, Peer-Sum, MixSet	70,000
Overall	All sources combined	680,767

Tableau A.2 : French Dataset sources

Split	Sources	Total
Train	HC3-OOD, French Instruct Dataset	464,885
Test	HC3-OOD (independent), French Instruct Dataset (independent)	15,000
Overall	All sources combined	479,885

Table A.1 and Table A.2 summarize the composition of the English and French datasets used in this study. For each language, the tables present the dataset split, the corresponding data sources, and the total number of samples. The English dataset combines multiple heterogeneous corpora, including HC3, M4GT, and MAGE for training, along with several independent benchmark datasets for testing. Similarly, the French dataset is constructed using HC3-OOD and the French Instruct Dataset, with separate independent subsets reserved for evaluation. Overall, these datasets provide a diverse multilingual benchmark for evaluating the robustness and generalization capability of AI-generated text detection models across different domains and generative sources.

B. Appendix B – Distribution of Human and AI-Generated Samples :

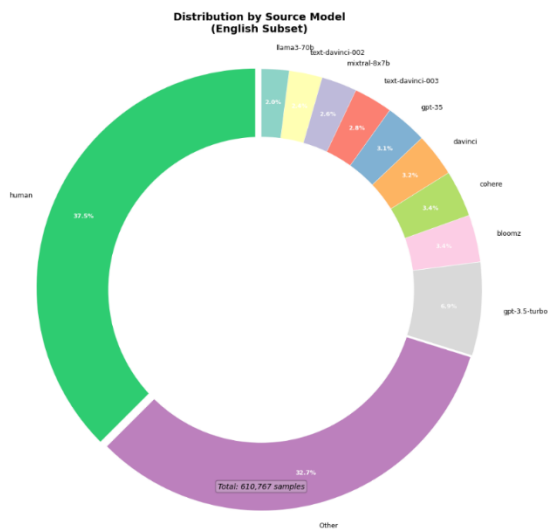


Figure B.B.1 : Distribution of Dataset Samples by Source

C. Appendix C – Text length distributions :

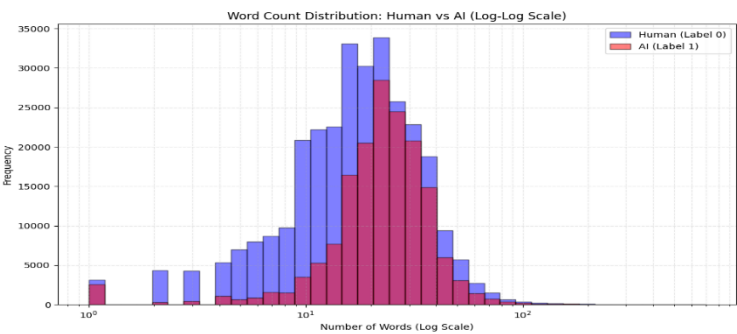


Figure C.1 : English text distributions in log scale

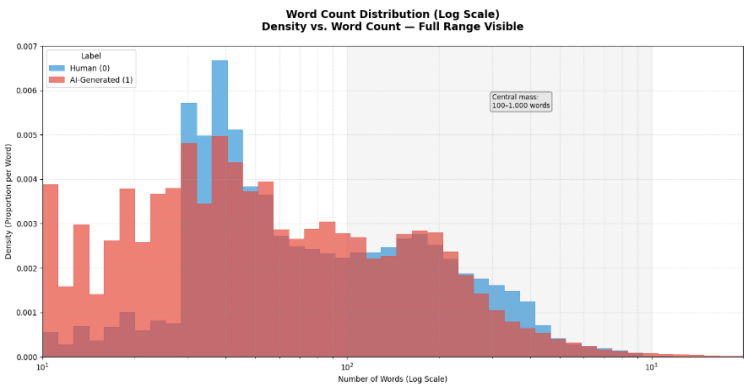


Figure C.2 : French text distributions in log scale

Figures C.1 and C.2 represent the logarithmic relationship between word frequency and occurrence count for the English and French datasets, respectively. These visualizations illustrate the distribution of token frequencies within each corpus and provide insights into the lexical characteristics of the datasets. The observed distributions follow a typical long-tail behavior, where a small number of words occur very frequently while the majority appear relatively rarely.

D. Appendix D – Additional Threshold Optimization Analysis (Case mDistil BERT with LoRA) :

Figure D.1 illustrates the threshold optimization analysis performed on the validation set. The figure combines the Precision–Recall (PR) curve, the score-threshold relationship, and the F1-score evolution across different threshold values. These visualizations provide insight into the trade-off between precision and recall as the decision threshold changes and help identify the optimal operating point used for final classification. The selected threshold corresponds to the configuration achieving the best balance between detection performance and classification reliability.

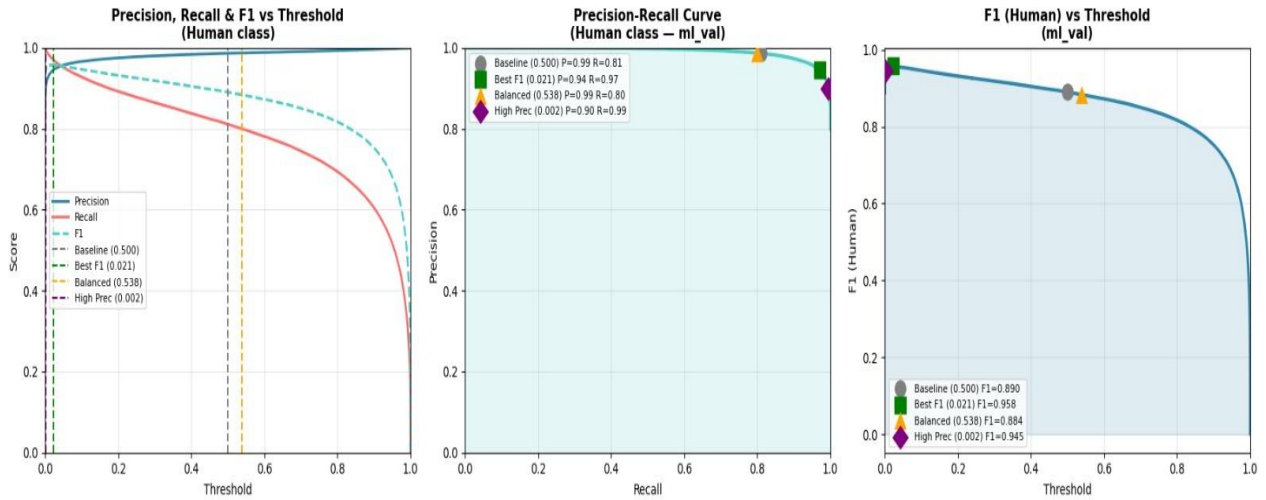


Figure D.1 : Score-threshold curve + PR curve + F1 threshold curve

E. Appendix E – Model Architectures :

Electra :

ELECTRA introduces a more efficient pre-training strategy for transformer language models compared to traditional masked language modeling approaches such as BERT. Instead of predicting masked words directly, ELECTRA trains the model to identify whether each token in a sentence is original or has been replaced.

The architecture is composed of two components: a small generator and a discriminator. First, some words in the input sentence are masked and replaced by the generator with predicted tokens. The discriminator then analyzes the modified sentence and determines for each token whether it is original or generated. Through this process, the model learns richer contextual representations by examining

every token in the sequence rather than only masked positions.

Unlike Generative Adversarial Networks (GANs), the generator is not trained adversarially. Instead, it is trained using standard language modeling objectives, while the discriminator learns to distinguish real tokens from replaced ones. After pre-training, the generator is discarded and only the discriminator is retained for downstream tasks such as text classification.

This training strategy improves computational efficiency and allows ELECTRA to achieve strong language understanding performance

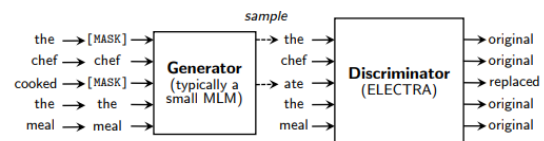


Figure E.1 : An overview of replaced token review strategy.

with fewer training resources compared to traditional transformer models.

efficiency and cross-lingual generalization are important.

Tableau E.1 : Electra Layers

Hyperparameter	Small	Base	Large
Number of layers	12	12	24
Hidden Size	256	768	1024
FFN inner hidden size	1024	3072	4096
Attention heads	4	12	16
Attention head size	64	64	64
Embedding Size	128	768	1024
Generator Size (multiplier for hidden-size, FFN-size, and num-attention-heads)	1/4	1/3	1/4

mBERT-Distill :

mBERT-Distil is a lightweight version of the multilingual BERT model designed to reduce computational cost while preserving most of its language understanding capabilities. It is obtained through a process called knowledge distillation, where a large pretrained multilingual BERT model (teacher) transfers its knowledge to a smaller model (student). The architecture follows the standard Transformer encoder design used in BERT, where input tokens are converted into contextualized representations through multiple self-attention layers. However, in mBERT-Distil, the number of layers is reduced, making the model faster and more efficient while still capturing important multilingual semantic information. Despite being smaller, mBERT-Distil retains strong performance across multiple languages because it is trained on large-scale multilingual corpora. This makes it well-suited for tasks such as text classification, where both

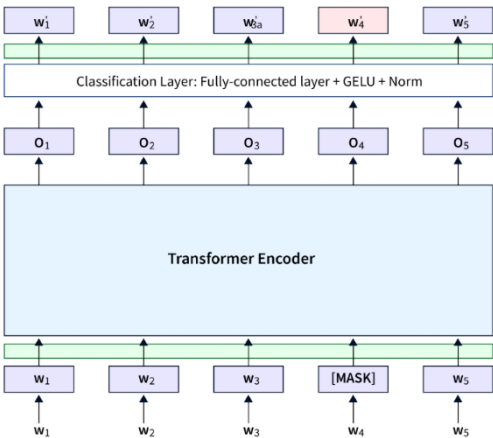


Figure E.2 : An overview of a transformer encoder classifier

LoRA (Low-Rank Adaptation)

LoRA is a parameter-efficient fine-tuning technique designed to adapt large pre-trained language models without updating all of their parameters. Instead of modifying the full model, LoRA introduces small trainable low-rank matrices into specific components of the Transformer architecture, typically within the attention layers. During fine-tuning, the original model weights are kept frozen, and only these additional low-rank matrices are trained. This significantly reduces the number of trainable parameters, making the training process more efficient in terms of memory and computation while still preserving strong performance.

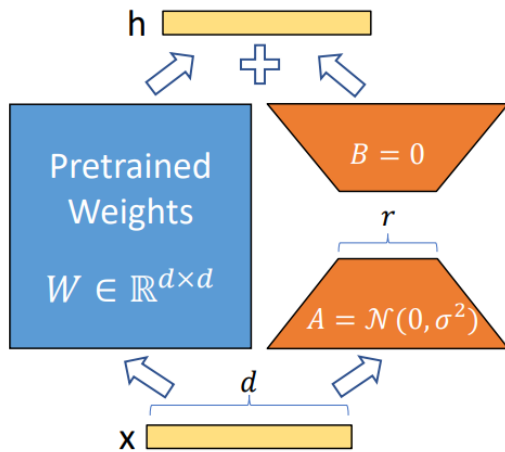


Figure E.3 : Visual Illustration of LoRA

In practice, LoRA allows large models such as mDistilBERT to be adapted to downstream tasks like text classification with minimal computational cost. This makes it particularly suitable for resource-constrained settings while maintaining competitive accuracy. In this work, LoRA is applied to mDistilBERT for the task of distinguishing between human-written and AI-generated text in a multilingual setting.

F. Appendix F — Evaluation Metrics

To evaluate model performance, we use standard classification metrics: accuracy, precision, recall, and F1-score. These metrics are computed

using true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN).

$$\begin{aligned} \text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN} \\ \text{Precision} &= \frac{TP}{TP + FP} \\ \text{Recall} &= \frac{TP}{TP + FN} \end{aligned}$$

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

These metrics are reported per class (Human and AI) as well as using macro-averaged scores to ensure balanced evaluation under potential class imbalance.